

Modal Preliminaries

Chapter 1

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The modal language

We work in a modal language $\mathcal{L}^M$ with propositional variables $\text{Prop} = \{p, q, r, \dots\}$, the Boolean connectives, and a unary necessity operator $\Box$. Its dual is

$$\Diamond A := \neg \Box \neg A \quad (\text{"possibly } A\text{"}).$$

Definition 1 (Frame)

A *Kripke frame* is a pair $\mathfrak{F}_R = \langle W, R \rangle$ with $W \neq \emptyset$ and $R \subseteq W \times W$.

Definition 2 (Model)

A *Kripke model* $\mathfrak{M}_R = \langle W, R, V \rangle$ adds a valuation $V : \text{Prop} \rightarrow \wp(W)$.

The modal clause:

$$\mathfrak{M}_R, w \Vdash \Box A \quad \text{iff} \quad \text{for all } v, wRv \text{ implies } \mathfrak{M}_R, v \Vdash A.$$

The smallest normal modal logic **K** extends classical propositional logic with

- the axiom **K**: $\Box(A \rightarrow B) \rightarrow (\Box A \rightarrow \Box B)$;
- the rule Nec: from $\vdash A$, infer $\vdash \Box A$.

Familiar extensions include **KT** = **K** $\oplus$ **T** (with **T** the schema $\Box A \rightarrow A$) and **S4**.

Theorem 3 (Completeness)

***K** is sound and complete with respect to the class of all Kripke frames [1, ch. 4].*

- [1] Patrick Blackburn, Maarten de Rijke, and Yde Venema. *Modal Logic*. Vol. 53. Cambridge Tracts in Theoretical Computer Science. Cambridge: Cambridge University Press, 2001. ISBN: 978-0521527149. DOI: 10.1017/CB09781107050884.